

GPT and BERT

DATA 443/643 · Week 3, note 3.4

Week 3

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Ideas from Princeton COS 484 L10 and CS224N pretraining (GPT vs BERT). Original slides; those courses are not copied.

Lecture plan

- 1 Static table versus contextual vector

- 2 Causal next-token versus masked tokens

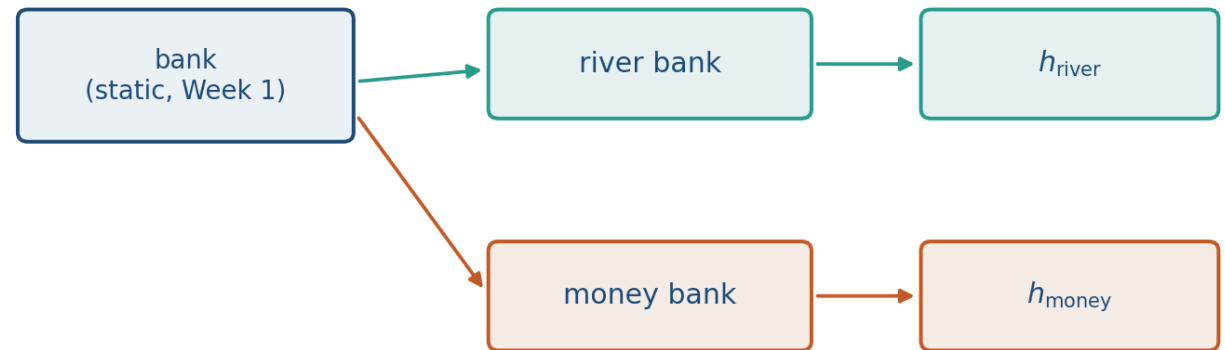
- 3 Same width, different mask and loss

- 4 What a project is allowed to stream

The same type, two vectors

- Week 1: bank is one row.
- A transformer: river bank is not money bank.
- Pretrain once, fine-tune many times.

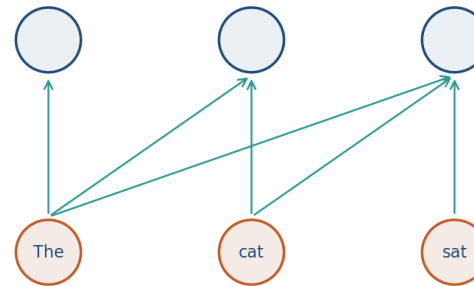
A transformer embedding is contextual: the same type, two vectors



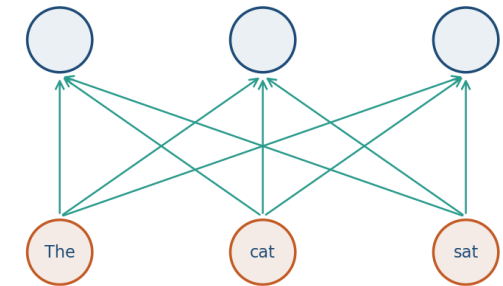
Same block, different mask

- GPT: look left, predict the next token.
- BERT: look both ways, fill [MASK].
- Those are two different uses of the word mask.

GPT: causal (next token)



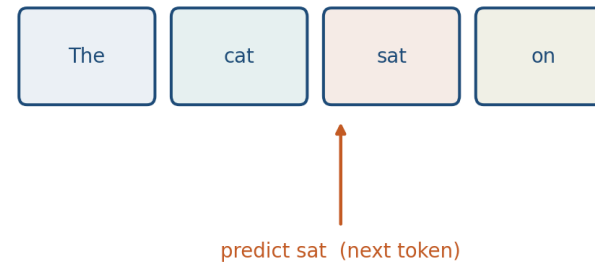
BERT: bidirectional (masked token)



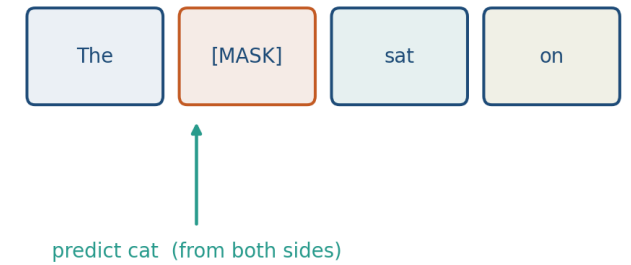
Two objectives

- GPT: $p(x_t \mid x_{1:t-1})$.
- BERT: $p(\text{masked token} \mid \text{rest})$.
- Do not fine-tune BERT as if it were GPT.

GPT: causal language model



BERT: masked language model



What you can ship

Decoder (GPT-style)

Generate, chat, RAG, tools. The default stack for this course.

Encoder (BERT-style)

Classify, tag, embed a span. Cannot stream a paragraph.

A causal 2 by 2

$$S_{\text{GPT}} = \begin{bmatrix} 1 & -\infty \\ 3 & 4 \end{bmatrix}$$

Row 1 softmax is [1, 0]. Token 1 cannot see token 2.

Row 2 is softmax([3, 4]) about [0.27, 0.73].

BERT softmaxes the unmasked S. Token 1 does look ahead.

What to take from this lecture

Context

Embeddings are now a function of the sentence.

Fork

Causal LM versus MLM. Same d , different job.

Project

If it must stream tokens, it is not BERT.